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### Lens retaining device and projection display apparatus

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#### Abstract

A lens retaining device includes a first shifter and a second shifter. The first shifter holds a lens and moves in a first direction perpendicular to an optical axis direction of the lens. The second shifter supports the first shifter and moves in a second direction perpendicular to the optical axis direction and the first direction. The first shifter includes a first main body moves in the first direction, a lens retainer that is provided on the first main body and retains the lens, and a first drive portion that moves the first main body. The lens retainer is integrally formed with the first main body.

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**Background/Summary**

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a lens retaining device and a projection display apparatus.

2. Description of the Related Art

(2) Patent Literature (PTL) 1 discloses a projector. The projector described in PTL 1 includes a lens shift mechanism that can move a projection lens in a direction perpendicular to the optical axis direction. PTL 1 is Unexamined Japanese Patent Publication No. 2015-14666.

SUMMARY

(3) However, the device described in PTL 1 still has room for improvement in terms of improvement in lens position accuracy.

(4) An object of the present disclosure is to provide a lens retaining device and a projection display

apparatus with improved lens position accuracy.

(5) A lens retaining device according to one aspect of the present disclosure includes: a first shifter that holds a lens and moves in a first direction perpendicular to an optical axis direction of the lens; and a second shifter that supports the first shifter and moves in a second direction perpendicular to the optical axis direction and the first direction. The first shifter includes: a first main body moves in the first direction; a lens retainer that is provided on the first main body and retains the lens; and a first drive portion that moves the first main body. The lens retainer is integrally formed with the first main body.

(6) A projection display apparatus according to one aspect of the present disclosure includes the lens retaining device according to the above aspect.

(7) The present disclosure can provide a lens retaining device and a projection display apparatus with improved lens position accuracy.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a schematic configuration of a projection display apparatus according to a first exemplary embodiment;

(2) FIG. 2 is a diagram illustrating a lens retaining device to which a projection lens is attached;

(3) FIG. 3 is a diagram illustrating the lens retaining device from which the projection lens is removed;

(4) FIG. 4 is a perspective view of the lens retaining device according to the first exemplary embodiment;

(5) FIG. 5 is a schematic exploded view of the lens retaining device of FIG. 4;

(6) FIG. 6 is a cross-sectional view of the lens retaining device of FIG. 4 taken along line VI-VI; and

(7) FIG. 7 is a cross-sectional view of the lens retaining device of FIG. 4 taken along line VII-VII.

### DETAILED DESCRIPTIONS

(8) Hereinafter, an exemplary embodiment will be described in detail with reference to the drawings as appropriate. However, unnecessarily detailed descriptions may be omitted. For example, a detailed description on an already well-known matter and a redundant description on the substantially same configuration may be omitted. This is to avoid the following descriptions to be unnecessarily redundant and to facilitate those skilled in the art to understand the following descriptions.

(9) Note that, the accompanying drawings and the following descriptions are provided to facilitate those skilled in the art to sufficiently understand the present disclosure, and are not intended to limit the subject matter described in the claims.

#### First Exemplary Embodiment

(10) [1-1. Configuration of Projection Display Apparatus]

(11) FIG. 1 is a block diagram illustrating a schematic configuration of projection display apparatus **100** according to a first exemplary embodiment. As illustrated in FIG. 1, projection display apparatus **100** includes lens retaining device **1**, projection lens **2**, optical unit **3**, and controller **4**. Lens retaining device **1**, optical unit **3**, and controller **4** are housed in housing **5**. Note that, projection display apparatus **100** projects an image onto a projection target such as a screen or a building.

(12) Lens retaining device **1** retains projection lens **2** and moves projection lens **2**. Lens retaining device **1** includes a retaining mechanism that retains projection lens **2** and a shift mechanism that moves projection lens **2**. In lens retaining device **1**, the retaining mechanism and the shift mechanism are integrally configured.

(13) FIG. 2 illustrates lens retaining device 1 to which projection lens 2 is attached, and FIG. 3 illustrates lens retaining device 1 from which projection lens 2 is removed. Note that, in FIGS. 2 and 3, components of projection display apparatus 100 other than lens retaining device 1 and projection lens 2 are not shown for ease of description. Further, X, Y, and Z-directions in the drawing respectively indicate a lateral direction, a thickness direction, and a vertical direction of lens retaining device 1.

(14) As illustrated in FIGS. 2 and 3, the retaining mechanism detachably retains an end side of projection lens 2. Specifically, the retaining mechanism retains projection lens 2 by engaging with engagement portion 2a provided at an end of projection lens 2. Further, projection lens 2 is removed from the retaining mechanism by releasing the engagement between the retaining mechanism and engagement portion 2a of projection lens 2. Engagement portion 2a is configured with, for example, a plurality of plate-shaped protrusions protruding outward at a peripheral edge of projection lens 2. The plurality of protrusions are provided at equal intervals on the outer periphery of projection lens 2.

(15) The shift mechanism moves projection lens 2 while retaining projection lens 2. Specifically, the shift mechanism moves projection lens 2 in a first direction D1 and a second direction D2 that are perpendicular to a direction in which optical axis P0 of projection lens 2 extends (hereinafter, referred to as an “optical axis direction”). In the present exemplary embodiment, the optical axis direction represents the Y-direction, the first direction D1 represents the X-direction (horizontal direction), and the second direction D2 represents the Z-direction (vertical direction).

(16) Projection lens 2 is a lens that enlarges image light and projects the image light onto a projection target. Projection lens 2 has a cylindrical shape having a first end and a second end. Projection lens 2 has engagement portion 2a provided on the second end side, and engagement portion 2a engages with lens retaining device 1.

(17) Optical unit 3 includes a light source device and a light modulation device. The light source device is a device that radiates light such as semiconductor laser. The light modulation device is, for example, a liquid crystal display element or a digital micromirror device (DMD).

(18) Controller 4 controls lens retaining device 1 and optical unit 3. Controller 4 can be implemented by a semiconductor device or the like. Controller 4 may be configured with, for example, a microcomputer, a central processing unit (CPU), a microprocessor unit (MPU), a graphics processor unit (GPU), a digital signal processor (DSP), a field-programmable gate array (FPGA), or an application specific integrated circuit (ASIC). A function of controller 4 may be configured only with hardware, or may be implemented by a combination of hardware and software. Controller 4 reads out data and a program stored in a storage such as a memory to perform various types of arithmetic processing and thus implements a predetermined function.

(19) [1-2. Configuration of Lens Retaining Device]

(20) FIG. 4 is a perspective view of lens retaining device 1 according to the first exemplary embodiment. FIG. 5 is a schematic exploded view of the lens retaining device of FIG. 4. FIG. 5 illustrate main component parts of lens retaining device 1, and some elements are not illustrated.

(21) As illustrated in FIGS. 4 and 5, lens retaining device 1 includes first shifter 10, second shifter 20, and base 30. First shifter 10 is supported by second shifter 20, and second shifter 20 is supported by base 30. In addition, lens retaining device 1 is provided with through hole 6 passing through first shifter 10, second shifter 20, and base 30. Image light passes through through hole 6.

(22) First shifter 10 retains projection lens 2 and moves in the first direction D1 perpendicular to the optical axis direction of projection lens 2. First shifter 10 is supported by second shifter 20 movably in the first direction D1. First shifter 10 includes: the retaining mechanism that retains projection lens 2; and the shift mechanism that moves in the first direction D1 perpendicular to the optical axis direction of projection lens 2. First shifter 10 integrally constitutes the retaining mechanism and the shift mechanism. Specifically, first shifter 10 includes first main body 11, lens retainer 12, first drive portion 13, and first shafts 14, wherein lens retainer 12 and first main body

**11** are integrally provided.

(23) In the present exemplary embodiment, first shafts **14** are fixed to second shifter **20**, and function as guides when first main body **11** moves. That is, first main body **11** is moved along first shafts **14** by first drive portion **13**.

(24) First main body **11** and lens retainer **12** are made of, for example, a metal material. That is, first main body **11** and lens retainer **12** are integrally formed of a metal material. The metal material includes, for example, an aluminum alloy or a magnesium alloy. In other words, first main body **11** and lens retainer **12** may be integrally formed of an aluminum alloy. Alternatively, first main body **11** and lens retainer **12** may be integrally formed of a magnesium alloy.

(25) First main body **11** is a plate-shaped member movable in the first direction **D1**. First main body **11** has, for example, a rectangular shape when viewed in the optical axis direction. In addition, first through hole **6a** having a circular shape is provided at the center of first main body **11** when viewed in the optical axis direction.

(26) First main body **11** includes first shaft retainers **16** each disposed along the first direction **D1** and provided with first shaft hole **15**. In the present exemplary embodiment, first main body **11** includes four first shaft retainers **16**. Four first shaft retainers **16** are provided at both ends of first main body **11** in the vertical direction (**Z**-direction). Specifically, two of first shaft retainers **16** are provided at a first end of first main body **11** in the vertical direction (**Z**-direction), and two of first shaft retainers **16** are provided at a second end of first main body **11** in the vertical direction (**Z**-direction).

(27) First shaft holes **15** are through holes having a circular shape when viewed in the first direction **D1**. First shafts **14** are disposed in first shaft holes **15**. In the present exemplary embodiment, first bearing **16a** is disposed in each of first shaft holes **15**. Each of first bearings **16a** is disposed on first shaft retainer **16** in one of first shaft holes **15**. First shafts **14** are disposed in first bearings **16a**.

(28) First shaft retainers **16** are provided to protrude from the ends of first main body **11** in the vertical direction (**Z**-direction). On first shaft retainers **16** that are connected to first drive portion **13** and are disposed on the first end side in the vertical direction (**Z**-direction) of first main body **11**, there is provided first connection portion **16b** that is connected to first drive portion **13**. First connection portion **16b** is a plate-shaped member protruding from first shaft retainers **16**. Two first shaft retainers **16** disposed on one end side of first main body **11** in the vertical direction (**Z**-direction) are arranged at an interval therebetween in the lateral direction (**X**-direction) of first main body **11**.

(29) Lens retainer **12** retains projection lens **2**. Lens retainer **12** retains projection lens **2** by engaging with engagement portion **2a** provided at the end side of projection lens **2**. Lens retainer **12** is provided on first main body **11**. Specifically, lens retainer **12** is provided on an inner wall that defines first through hole **6a** provided in first main body **11**.

(30) Lens retainer **12** has a plurality of ribs **12a** having a plate shape protruding from the inner wall of first through hole **6a** toward the inside of first main body **11**. The plurality of ribs **12a** are provided at equal intervals when viewed in the optical axis direction. In the present exemplary embodiment, the plurality of ribs **12a** include three ribs **12a**.

(31) The plurality of ribs **12a** are provided on the side of first main body **11** where second shifter **20** is disposed. Each of the plurality of ribs **12a** has contact surface **12b** in contact with projection lens **2**. Contact surfaces **12b** are perpendicular to the optical axis direction of projection lens **2** and are in contact with engagement portion **2a** of projection lens **2**.

(32) In addition, lens retainer **12** includes a plurality of locking portions **12c**. The plurality of locking portions **12c** are provided at equal intervals when viewed in the optical axis direction. In the present exemplary embodiment, the plurality of locking portions **12c** includes three locking portions **12c**.

(33) The plurality of locking portions **12c** lock engagement portions **2a**, of projection lens **2**,

disposed on contact surfaces **12b** of plurality of ribs **12a**. In the present exemplary embodiment, engagement portion **2a** is formed of a plurality of protrusions. The plurality of locking portions **12c** lock the plurality of protrusions disposed on contact surfaces **12b** of the plurality of ribs **12a**. To attach projection lens **2** to lens retainer **12**, engagement portion **2a** of projection lens **2**, that is, the plurality of protrusions are inserted into first through hole **6a**, so that engagement portion **2a** of projection lens **2** is brought into contact with contact surfaces **12b** of the plurality of ribs **12a**. Next, projection lens **2** is rotated about the optical axis to slidably move engagement portion **2a** on contact surfaces **12b**, thereby engaging engagement portion **2a** with the plurality of locking portions **12c**. As a result, engagement portion **2a** of projection lens **2**, that is, the plurality of protrusions are locked and fixed by the plurality of locking portions **12c**. To detach projection lens **2**, a procedure reverse to the operation of attaching projection lens **2** is performed.

(34) First drive portion **13** moves first main body **11** in the first direction **D1**. First drive portion **13** has first actuator **17** that applies force to move first main body **11** in the first direction **D1**.

(35) First actuator **17** moves first main body **11** along first shafts **14** in the first direction **D1** by applying force to first main body **11**. As first actuator **17**, a motor or the like can be used, for example. First actuator **17** is connected to first connection portion **16b** of first shaft retainers **16** provided on first main body **11**. First actuator **17** applies force to first connection portion **16b** in the first direction **D1**. With this arrangement, first main body **11** moves in the first direction **D1**, being guided by first shafts **14**. First actuator **17** is connected to first connection portion **16b** via, for example, a ball screw mechanism or a rotation transmission gear mechanism. The ball screw mechanism is, for example, a mechanism having a screw portion and a nut portion. The rotation transmission gear mechanism is, for example, a mechanism in which a plurality of gears are combined. First actuator **17** is controlled by, for example, controller **4**.

(36) First shafts **14** are columnar shafts disposed in first shaft holes **15** of first shaft retainers **16**. First shafts **14** extend in the first direction **D1**. First shafts **14** are disposed in first bearings **16a** and are disposed in first shaft holes **15** via first bearings **16a**. First shafts **14** are slidably supported on first bearings **16a** by first shaft retainers **16**. First shafts **14** are fixed to second shifter **20** and function as a guide when first main body **11** moves. Since first shifter **10** moves, being guided by first shafts **14** fixed to second shifter **20**, first shifter **10** moves in the first direction **D1** with respect to second shifter **20**. Therefore, even when first shifter **10** moves in the first direction **D1**, second shifter **20** does not move in the first direction **D1**.

(37) <Second Shifter>

(38) Second shifter **20** supports first shifter **10** and moves in the second direction **D2** that is perpendicular to the optical axis direction and the first direction **D1**. Second shifter **20** supports first shifter **10** to be movable in the first direction **D1**. Further, second shifter **20** is supported by base **30** movably in the second direction **D2**. Therefore, second shifter **20** moves in the second direction **D2** together with first shifter **10** with respect to base **30**. Second shifter **20** includes a second main body **21**, second drive portion **22**, and second shafts **23**.

(39) Second main body **21** is a plate-shaped member movable in the second direction **D2**. Second main body **21** has, for example, a rectangular shape when viewed in the optical axis direction. In addition, at the center of second main body **21** when viewed in the optical axis direction, there is provided second through hole **6b** having a circular shape. Second through hole **6b** is provided at a position overlapping, when viewed in the optical axis direction, first through hole **6a** and communicates with first through hole **6a**.

(40) Second main body **21** includes second shaft retainers **25** each disposed along the second direction **D2** and provided with second shaft hole **24**. In the present exemplary embodiment, second main body **21** includes four second shaft retainers **25**. Four second shaft retainers **25** are provided at both ends of second main body **21** in the lateral direction (X-direction). Specifically, two of second shaft retainers **25** are provided at a first end of second main body **21** in the lateral direction (X-direction), and two of second shaft retainer **25** are provided at a second end of second

main body **21** in the lateral direction (X-direction).

(41) Second shaft holes **24** are through holes having a circular shape when viewed in the second direction **D2**. Second shafts **23** are disposed in second shaft holes **24**. In the present exemplary embodiment, second bearing **25a** is disposed in each second shaft hole **24**. Second bearing **25a** is disposed on second shaft retainer **25** in one of second shaft holes **24**. Second shafts **23** are disposed in second bearings **25a**.

(42) Second shaft retainers **25** are provided to protrude from the ends of second main body **21** in the lateral direction (X-direction). On second shaft retainers **25** that are connected to second drive portion **22** and are disposed on the first end side in the lateral direction (X-direction) of second main body **21**, there is provided second connection portion **25b** connected to second drive portion **22**. Second connection portion **25b** is a plate-shaped member protruding from second shaft retainers **25**. Two second shaft retainers **25** disposed on one end side of second main body **21** in the lateral direction (X-direction) are arranged at an interval therebetween in the vertical direction (Z-direction) of second main body **21**.

(43) Second drive portion **22** moves second main body **21** in the second direction **D2**. Second drive portion **22** has second actuator **26** that applies force to move second main body **21** in the second direction **D2**.

(44) Second actuator **26** moves second main body **21** along second shafts **23** in the second direction **D2** by applying force to second main body **21**. Second actuator **26** may be similar to or different from first actuator **17**. Second actuator **26** is connected to second connection portion **25b** of second shaft retainers **25** provided on second main body **21**. Second actuator **26** applies force to second connection portion **25b** in the second direction **D2**. With this arrangement, second main body **21** moves in the second direction **D2**, being guided by second shafts **23**. Second actuator **26** is controlled by, for example, controller **4**.

(45) Second shafts **23** are columnar shafts disposed in second shaft holes **24** of second shaft retainers **25**. Second shafts **23** extend in the second direction **D2**. Second shafts **23** are disposed in second bearings **25a** and are disposed in second shaft holes **24** via second bearings **25a**. Second shafts **23** are slidably supported on second bearings **25a** by second shaft retainer **25**. In the present exemplary embodiment, second shafts **23** are fixed to base **30** and function as guides when second main body **21** moves. Since second shifter **20** moves, being guided by second shafts **23** fixed to base **30**, second shifter **20** moves in the second direction **D2** with respect to base **30**. First shifter **10** is fixed to second main body **21** via first shafts **14**. Therefore, when second main body **21** moves in the second direction **D2**, first shifter **10** also moves in the second direction **D2** together with second main body **21**.

(46) <Base>

(47) Base **30** supports second shifter **20**. Base **30** supports second shifter **20** to be movable in the second direction **D2**. Base **30** includes base main body **31**.

(48) Base main body **31** is a plate-shaped member. Base main body **31** has, for example, a rectangular shape when viewed in the optical axis direction. In addition, third through hole **6c** having a circular shape is provided at the center of base main body **31** when viewed in the optical axis direction. Third through hole **6c** is provided at a position overlapping, when viewed in the optical axis direction, first through hole **6a** and second through hole **6b**, and communicates with first through hole **6a** and second through hole **6b**. First through hole **6a**, second through hole **6b**, and third through hole **6c** form through hole **6** of lens retaining device **1**.

(49) FIG. **6** is a cross-sectional view of lens retaining device **1** of FIG. **4** taken along line VI-VI. As illustrated in FIG. **6**, contact surfaces **12b** of the plurality of ribs **12a** of lens retainer **12** are disposed at positions close to shaft centers **C1** of first shafts **14** in the optical axis direction (Y-direction). Shaft centers **C1** of first shafts **14** mean the centers of circular first shafts **14** when viewed in the first direction **D1** (X-direction).

(50) Here, when viewed in the first direction **D1** (X-direction), a plane perpendicular to the optical

axis direction and including shaft centers **C1** of first shafts **14** is defined as first virtual plane **PL1**. In the present exemplary embodiment, first virtual plane **PL1** is an XZ-plane passing through shaft centers **C1** of two first shafts **14**. A first distance **L1** between first virtual plane **PL1** and contact surfaces **12b** is smaller than a diameter **d1** of first shafts **14**. Preferably, the first distance **L1** is smaller than  $\frac{1}{2}$  times the diameter **d1** of first shafts **14**. More preferably, the first distance **L1** is smaller than  $\frac{1}{3}$  times the diameter **d1** of first shafts **14**. The first distance **L1** means a distance between first virtual plane **PL1** and contact surfaces **12b** in the optical axis direction (Y-direction). (51) FIG. 7 is a cross-sectional view of lens retaining device **1** of FIG. 4 taken along line VII-VII. As illustrated in FIG. 7, contact surfaces **12b** of the plurality of ribs **12a** of lens retainer **12** are disposed at positions close to shaft centers **C2** of second shafts **23** in the optical axis direction (Y-direction). Shaft centers **C2** of second shafts **23** mean the centers of circular second shafts **23** when viewed in the second direction **D2** (Z-direction).

(52) Here, when viewed in the second direction **D2** (Z-direction), a plane perpendicular to the optical axis direction and including shaft centers **C2** of second shafts **23** is defined as second virtual plane **PL2**. In the present exemplary embodiment, second virtual plane **PL2** is an XZ-plane passing through shaft centers **C2** of two second shafts **23**. A second distance **L2** between second virtual plane **PL2** and contact surfaces **12b** is smaller than twice a diameter **d2** of second shafts **23**. Preferably, the second distance **L2** is smaller than  $\frac{3}{2}$  times the diameter **d2** of second shafts **23**. More preferably, the second distance **L2** is smaller than the diameter **d2** of second shafts **23**. The second distance **L2** means a distance between second virtual plane **PL2** and contact surfaces **12b** in the optical axis direction (Y-direction).

(53) [1-3. Advantageous Effects]

(54) Lens retaining device **1** of the present disclosure includes: first shifter **10** movable in the first direction **D1**; and second shifter **20** that supports first shifter **10** and is movable in the second direction **D2**. First shifter **10** includes: first main body **11** movable in the first direction **D1**; lens retainer **12** that is provided on first main body **11** and retains projection lens **2**; and first drive portion **13** that moves first main body **11**. Lens retainer **12** is formed integrally with first main body **11**. Such a configuration makes it possible to improve positional accuracy of projection lens **2**. Lens retaining device **1** of the present disclosure can improve mechanical strength as compared with a case where first main body **11** and lens retainer **12** are formed as separate members. Since the number of components and the number of manufacturing steps can be reduced, the manufacturing cost can be reduced. Since lens retainer **12** is formed integrally with first main body **11**, dimensional accuracy can be improved.

(55) First main body **11** and lens retainer **12** are made of a metal material. For example, as the metal material, an aluminum alloy or a magnesium alloy can be used. Such a configuration makes it possible to further improve the mechanical strength of lens retainer **12** as compared with a lens retainer made of a plastic material. By using a metal material, which has a lower thermal expansion coefficient than plastic materials, it is possible to improve resistance to environmental factors and temperature changes. Further, metal materials are less likely to deteriorate than plastic materials, and the positional accuracy and focus of projection lens **2** can be improved.

(56) In lens retaining device **1**, when viewed in the first direction **D1**, the first distance **L1**, which is between first virtual plane **PL1** perpendicular to the optical axis direction and including shaft centers **C1** of first shafts **14** and contact surfaces **12b** of lens retainer **12**, is smaller than the diameter **d1** of first shafts **14**. Such a configuration makes it possible to dispose first shafts **14** near contact surfaces **12b** of lens retainer **12**. Contact surfaces **12b** are surfaces that are in contact with the end surface on the second end side of projection lens **2** to support projection lens **2**. Projection lens **2** is retained, in a cantilever state, on the second end on which projection lens **2** is in contact with contact surfaces **12b**. Since first main body **11** of first shifter **10** moves while being guided by first shafts **14**, the positional accuracy of projection lens **2** can be improved by disposing contact surfaces **12b** supporting projection lens **2** and first shafts **14** to be close to each other.



(57) In lens retaining device **1**, when viewed in the second direction **D2**, the second distance **L2**, which is between contact surfaces **12b** and second virtual plane **PL2** perpendicular to the optical axis direction and including shaft centers **C2** of second shafts **23**, is smaller than twice the diameter **d2** of second shafts **23**. This configuration makes it possible to dispose second shafts **23** near contact surfaces **12b** of lens retainer **12**. Since second main body **21** of second shifter **20** moves while being guided by second shafts **23**, the positional accuracy of projection lens **2** can be improved by disposing contact surfaces **12b** supporting projection lens **2** and second shafts **23** to be close to each other.

(58) A projection display apparatus including lens retaining device **1** has the same effect as lens retaining device **1**.

(59) Note that, the present exemplary embodiment has described an example in which lens retaining device **1** retains projection lens **2**, but the present disclosure is not limited to such an example. Lens retaining device **1** may be a device that retains a lens other than projection lens **2**.

(60) The present exemplary embodiment has described an example in which lens retaining device **1** includes base **30**, but the present disclosure is not limited to such an example. Base **30** is not an essential component.

(61) The present exemplary embodiment has described an example in which the first direction **D1** in which first shifter **10** moves is the X-direction and the second direction **D2** in which second shifter **20** moves is the Z-direction, but the present invention is not limited to such an example. For example, the first direction **D1** may be the Z-direction, and the second direction **D2** may be the X-direction.

(62) The present exemplary embodiment has described an example in which projection display apparatus **100** includes lens retaining device **1**, projection lens **2**, optical unit **3**, and controller **4**; however, the present disclosure is not limited to such an example. Projection display apparatus **100** may include elements other than these elements.

(63) As described above, the exemplary embodiment has been described to exemplify the techniques in the present disclosure. For this purpose, the accompanying drawings and the detailed description have been provided. Therefore, in order to illustrate the above techniques, the components described in the accompanying drawings and the detailed description can include not only the components necessary to solve the problem but also components unnecessary to solve the problem. For this reason, it should not be immediately recognized that those unnecessary components are necessary just because those unnecessary components are described in the accompanying drawings and the detailed description.

(64) The above exemplary embodiment is provided to exemplify the techniques according to the present disclosure. Therefore, it is possible to make various changes, replacements, additions, omissions, and the like, within the scope of the claims and equivalents thereof.

(65) (Outline of Exemplary Embodiment)

(66) Item (1): A lens retaining device of the present disclosure includes: a first shifter that holds a lens and moves in a first direction perpendicular to an optical axis direction of the lens; and a second shifter that supports the first shifter and moves in a second direction perpendicular to the optical axis direction and the first direction. The first shifter includes: a first main body moves in the first direction; a lens retainer that is provided on the first main body and retains the lens; and a first drive portion that moves the first main body. The lens retainer is integrally formed with the first main body.

(67) Item (2): In the lens retaining device of item (1), the first main body and the lens retainer may be made of a metal material.

(68) Item (3): In the lens retaining device of item (2), the metal material may contain an aluminum alloy or a magnesium alloy.

(69) Item (4): The lens retaining device of any one of items (1) to (3) may have the following configuration. The first shifter includes a first shaft that is fixed to the second shifter and guides

movement of the first main body; the first main body includes a first shaft retainer disposed along the first direction and provided with a first shaft hole that the first shaft is disposed in; the lens retainer has a contact surface that is perpendicular to the optical axis direction and is in contact with the lens; the first drive portion includes a first actuator that applies force to move the first main body in the first direction; and when viewed in the first direction, a first distance between the contact surface and a first virtual plane that is perpendicular to the optical axis direction and includes a shaft center of the first shaft is smaller than a diameter of the first shaft.

(70) Item (5): In the lens retaining device of item (4), the first distance may be smaller than  $\frac{1}{2}$  of the diameter of the first shaft.

(71) Item (6): In the lens retaining device of item (4) or (5), the second shifter may include: a second main body that moves in the second direction; and a second drive portion that moves the second main body.

(72) Item (7): The lens retaining device of item (6) may have the following configuration. The second shifter includes a second shaft that guides movement of the second main body; the second main body includes a second shaft retainer disposed along the second direction and provided with a second shaft hole that the second shaft is disposed in; the second drive portion includes a second actuator that applies force to move the second main body in the second direction; and when viewed in the second direction, a second distance between the contact surface and a second virtual plane that is perpendicular to the optical axis direction and includes a shaft center of the second shaft is smaller than twice a diameter of the second shaft.

(73) Item (8): The lens retaining device of any one of items (1) to (7) may further include a base that supports the second shifter.

(74) Item (9): A projection display apparatus according to the present disclosure includes the lens retaining device of any one of items (1) to (8).

(75) The present disclosure is applicable to a lens retaining device capable of adjusting a position of a lens, and is applicable to a projection display apparatus that projects a video.

## Claims

1. A lens retaining device comprising: a first shifter that holds a lens and moves in a first direction perpendicular to an optical axis direction of the lens; and a second shifter that supports the first shifter and moves in a second direction perpendicular to the optical axis direction and the first direction, wherein the first shifter includes: a first main body that moves in the first direction; a first shaft that is fixed to the second shifter and guides movement of the first main body; a lens retainer that is provided on the first main body and retains the lens; and a first drive portion that moves the first main body, the lens retainer is integrally formed with the first main body, the first main body includes a first shaft retainer disposed along the first direction and provided with a first shaft hole that the first shaft is disposed in, the lens retainer has a contact surface that is perpendicular to the optical axis direction and is in contact with the lens, the first drive portion includes a first actuator that applies force to move the first main body in the first direction, and when viewed in the first direction, a first distance between the contact surface and a first virtual plane that is perpendicular to the optical axis direction and includes a shaft center of the first shaft is smaller than a diameter of the first shaft.

2. The lens retaining device according to claim 1, wherein the first main body and the lens retainer are made of a metal material.

3. The lens retaining device according to claim 2, wherein the metal material contains an aluminum alloy or a magnesium alloy.

4. The lens retaining device according to claim 1, wherein the first distance is smaller than  $\frac{1}{2}$  of the diameter of the first shaft.

5. The lens retaining device according to claim 1, wherein the second shifter includes: a second

main body that moves in the second direction; and a second drive portion that moves the second main body.

6. The lens retaining device according to claim 5, wherein the second shifter includes a second shaft that guides movement of the second main body, the second main body includes a second shaft retainer disposed along the second direction and provided with a second shaft hole that the second shaft is disposed in, the second drive portion includes a second actuator that applies force to move the second main body in the second direction, and when viewed in the second direction, a second distance between the contact surface and a second virtual plane that is perpendicular to the optical axis direction and includes a shaft center of the second shaft is smaller than twice a diameter of the second shaft.

7. The lens retaining device according to claim 1, further comprising a base that supports the second shifter.

8. A projection display apparatus comprising the lens retaining device according to claim 1.

9. A lens retaining device comprising: a first shifter that holds a lens and moves in a first direction intersecting an optical axis direction of the lens; and a second shifter that supports the first shifter and moves in a second direction intersecting the optical axis direction and the first direction, wherein the first shifter includes: a first main body that moves in the first direction; a first shaft that is fixed to the second shifter and guides movement of the first main body; a lens retainer that is provided on the first main body and retains the lens; and a first drive portion that moves the first main body, the lens retainer is integrally formed with the first main body, the first main body includes a first shaft retainer disposed along the first direction and provided with a first shaft hole that the first shaft is disposed in, the lens retainer has a contact part that is perpendicular to the optical axis direction and is in contact with the lens, the first drive portion includes a first actuator that applies force to move the first main body in the first direction, and when viewed in the first direction, a first distance between the contact part and a first virtual plane that is perpendicular to the optical axis direction and includes a shaft center of the first shaft is smaller than a diameter of the first shaft.

10. A projection display apparatus comprising the lens retaining device according to claim 9.

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